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A dense multi-pooling convolutional network for driving fatigue detection

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Driver fatigue is one of the major causes of traffic accidents, particularly for drivers of large vehicles, who are more susceptible to fatigue due to prolonged driving hours and monotonous conditions during their journeys. Existing vision-based driver fatigue detection methods fail to accurately and timely judge fatigue in complex driving scenarios (e.g., when wearing glasses or in the presence of non-driver individuals). To address these issues, this paper proposes a driving fatigue detection method based on a novel network and the analysis of driver facial actions. The proposed approach mainly consists of three submodules, i.e. Driver's State Detection (DSD), Dense Multi-Pooling Convolutional Network (DMP-Net), and Driving Fatigue Detection (DFD). In the DSD module, MTCNN is employed to locate the driver's face and detect facial landmarks in real time. Additionally, a face detection bounding box filtering algorithm is proposed to reduce false detections of the driver. To accurately detect the states of the driver's facial actions, we propose the DMP-Net network, which contains only a small number of parameters and outperforms existing methods in terms of accuracy and time consumption. The DFD module determines whether the driver is fatigued by comparing a reasonable threshold with the frequency of mouth opening (FM) and the percentage of eyelid closure over the pupil over time parameter (PERCLOS). Results of the experiments based on benchmarks and our self-collected datasets show that our method achieves 99.25% accuracy on the CEW dataset, 99.24% accuracy on the ZJU dataset, and 99.12% accuracy on our self-collected dataset. Our proposed driving fatigue detection method has as a high accuracy in real time and outperforms the existing methods.

Intelligent driving including assisting driving and providing early warning signals is a hot topic in the field of artificial intelligence research[1]. The National Highway Traffic Safety Administration reported that over 30% of respondents had experienced fatigue while driving. According to the European Transport Safety Council, fatigued driving contributed to 10% to 20% of all traffic accidents[2]. Preventing driver drowsiness is significantly more challenging compared to preventing speeding and other visually evident traffic violations. As a result, traffic safety has been severely compromised. Prolonged driving combined with insufficient sleep can lead to relaxation of ocular muscles, reduced eye movement, frequent blinking, and even double vision[3]. Fatigued driving can cause drivers to respond slowly to emergency situations, impair their ability to judge vehicle speed and distance between cars, and significantly increase the likelihood of traffic accidents. Therefore, Several methods have been developed to detect driver fatigue[4–11] over the past decades to improve the safety of traffic and help drivers drive more safely. However, these methods have some limitations. Vehicle-based methods[9] are not robust to a variety of situations because it is hard to find typical driving metrics to use as a criterion. Methods based on physiological signals[10,11] rely on onboard physiological equipment to collect different physiological signals. These complicated devices are worn by drivers, which may be uncomfortable for them, and they are difficult to promote in actual production. Table 1 presents research works based on visual detection in recent years[4–8], which indicate that vision-based fatigue detection methods have overcome the limitations of vehicle-based and physiological signal-based methods. However, these methods still exhibit mediocre performance in terms of real-time capability and accuracy, primarily because driver-related features extracted from images are susceptible to image noise (non-driver elements), driver head movements, and facial obstructions (such as glasses and masks). Importantly, the time between a driver becoming clearly aware of their fatigued driving state and the occurrence of a severe

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Methods	Description of methods	Performance analysis
Sun et al.[4]	This method combines contextual features, facial features, and vehicle behavior features to improve the ability to recognize fatigue driving.	Time: 287ms Accuracy:95.1%
Mandal et al.[5]	This method proposes an eye openness estimation and fusion algorithm based on multi-model eye detections for both eyes, in order to obtain accurate and robust estimates of the driver's eye openness.	Running speed:20fps Accuracy:85.02%
Gu et al.[6]	This method proposes a multi-task hierarchical CNN model for face detection, eye and mouth state detection, as well as fatigue detection.	Time: 4.3ms Accuracy:98.02 %
Dong et al.[7]	This method extracts facial features of the driver to determine actions such as yawning, head posture, and eyes being open or closed, thereby assessing whether the driver is fatigued.	Time: 240ms Accuracy:91.0%
Ours	Our method proposes DSD, DMP-Net, and DFD to extract the state of eyes and mouth. It detects driver fatigue based on the Frequency of Mouth open and the PERCLOS.	Time:3.5ms Accuracy:99.16%

Table 1. Table of a comparative analysis between the fatigue driving detection methods.

accident (including death and permanent disability) can be extremely short. Therefore, the time efficiency of driver fatigue detection is crucial.

To address these issues, a novel framework for detecting driver fatigue is proposed in this paper. This method adopts efficient time management practices at every step, achieving high accuracy and real-time performance. Specifically, our method consists of three modules: Driver's State Detection (DSD), Dense Multi-Pooling Convolutional Network (DMP-Net), and Driving Fatigue Detection (DFD). DSD removes non-driver noise while detecting facial landmarks of the driver to isolate key regions that reflect driver fatigue. DMP-Net effectively detects the states of the eyes and mouth. DFD makes the final decision on whether the driver is fatigued.

The main contributions of this paper are as follows:

(1) We propose a real-time fatigue driving detection method, which mainly includes three submodules, i.e. Driver's State Detection (DSD), Dense Multi-Pooling Convolutional Network (DMP-Net), and Driving Fatigue Detection (DFD).
(2) A Dense Multi-Pooling convolutional neural network (DMP-Net) is proposed to effectively detect the state of the eyes and mouth. This network has the characteristics of real-time and high detection accuracy.

Related work

Existing driving fatigue detection methods can be classified into three categories, the Vehicle Information based methods, the Physiological Measures based methods and the Visual-based methods.

Vehicle Information based methods. Fatigued driving can lead to changes in driving behaviors, such as speed, acceleration, gear shifting, and steering wheel movements. These features can be adopted to detect driver fatigue[9]. Takei et al. first use steering wheel motion to estimate fatigue in driving[12]. Another work using wheel data to detect fatigue driving is presented in[13]. Vehicle states are used in[14] proposed by Zhong et al. They use vehicle states such as trace profile and steering angle to analyze whether the driver is in a state of fatigue. However, the main shortcoming of these vehicle-based methods is that it's hard to find standard driving parameters as a criterion. Because individual driving characteristics are different and they are highly dependent on road conditions. These factors make these methods not robust to various situations.

Physiological signals-based methods. When a driver enters a state of fatigue, some human physiological signals will undergo noticeable changes. These physiological signals include brain waves, the electrical activity along the scalp (EEG), the electrical activity of the heart (ECG), electrical activity associated with eye movements (EOG), and the electrical activity produced by skeletal muscles (EMG)[15]. Compared with vehicle information-based methods, the physiological signals-based methods show better detection performance. Because the changes in biological indicators are obvious. Fu et al.[16] proposed a dynamic driver fatigue detection model. This model is based on Hidden Markov Model (HMM) and estimate fatigue in a probabilistic way using various physiological and contextual information. Zhang et al.[17] presented a real-time method based on various entropy and complexity measures for detection and identification of driving fatigue. They adopt three physiological signals: electroencephalogram (EEG), electromyogram (EMG), and electrooculogram (EOG). Four feal-time features are extracted and applied for training and testing. However, in order to obtain various physiological signals, we must rely on onboard physiological devices. These devices are complex and drivers must wear these devices, which may cause drivers discomfort and they are hard to promote in actual production.

Visual-based methods. With the development of Artificial Intelligence and Computer Vision[18,19], visual-based methods are also being applied in many applications[20,21], such as driving fatigue detection. These methods are non-contact detection technology that does not interfere with the driver's normal driving. Generally, the HD camera is used for data collection. Visual-based methods can divide into two research directions:

State of eyes and mouth. The most obvious manifestation of fatigue on the face is the change in the state of the eyes and mouth. PERCLOS is one of the commonly used indicators for eye closure detection. Chang et al[22] restore an infrared camera to capture the pupil images to eliminate the effects of eye occlusion. These images are classified using an artificial neural network. PERCLOS is calculated as the total time that the drivers eyelids are closed more than 80%[23]. Li et al.[24] proposed the Active Appearance Model (AAM). This model is used to detect the face region and the position of the eyes. PERCLOS is adopted as the indicator to determine whether the driver is drowsy or not. Yawning is another important indicator of fatigue. Petropoulakis et al.[25] proposed a method that detected yawning by the variation of the aspect ratio of the mouth profile. They carried out a series of experiments to find the critical value of the aspect ratio.

Detection of fatigue facial expressions. When people feel physically or mentally fatigued, their facial expressions change, such as displaying a blank expression and droopy eyes. Therefore, some scholars are committed to extracting facial features of drivers and classifying these features to determine whether the drivers are in a state of fatigue. Zhang et al.[26] presented an algorithm based on the local binary pattern. With the development of deep learning, researchers found that using feature representation to detect fatigue can achieve better recognition performance. A Deep Drowsiness Detection (DDD) network is proposed in[27]. In DDD, three deep networks are used to reach global robustness against the background and environmental variations, as well as to detect local facial movements reliably. Zhao et al.[28] use facial dynamic fusion information and a deep belief network (DBN) to detect fatigue. The texture information of the eye and mouth is combined with landmark information as a feature vector, and then DBN is used for feature classification. The author adopts DBN because the characteristics of DBN are less than CNN, this improves the real-time performance of the model.

Vision-based technology is the main research direction of driving fatigue detection. Vision-based methods possess unique advantages because they can provide non-invasive visual information about the driver without being influenced by external factors. With the advancement of computer vision technology, vision-based driving fatigue detection technology will achieve better performance.

Method

As shown in Fig. 1, Driver State Detection (DSD) is the first component of our proposed method for detecting driver fatigue. This component includes driver face detection as well as segmentation of the eyes and mouth. To accomplish these operations, we employ a trained multi-task cascaded convolutional network (MTCNN)[29], which is used for face and keypoint detection. The second part is Parallel Detection of Eye and Mouth State module which uses our proposed Dense Multi-Pooling Convolutional Network (DMP-Net) to detect the state of eyes and mouth. Finally, in the third part, the Frequency of Mouth Open (FM) parameter and the PERCLOS parameter are used to evaluate Driving Fatigue Detection (DFD).

Driver's state detection

The important first step in visual-based driving fatigue detection methods is face detection. With the development of deep learning, many methods[30] can detect faces accurately and efficiently. The MTCNN model, as a high-precision real-time face detection and alignment technology based on Convolutional Neural Networks (CNNs), becomes the preferred network architecture for DSD.

MTCNN[29] can detect five face landmarks and the location of the face in real-time. MTCNN network can be divided into three sub-networks: Proposal Network (P-Net), Refine Network (R-Net), and Output Network (O-Net). P-Net is a fully convolutional network. P-Net takes the constructed image pyramid as input. The bounding box regression vectors for candidate facial windows are obtained and the bounding box regression vectors are used to calibrate candidates. The authors adopt non-maximum suppression (NMS)[31] to merge the highly overleap candidates. After the operation of the P-Net, the candidates are fed to R-Net. R-Net is a CNN that further rejects a large number of false candidates. The NMS is also used to merge candidates in this net. The function of the O-Net is similar to R-Net, but this net aims to identify face regions with more supervision. After the operation of O-Net, we will get the location of the face and five facial landmark positions. The structure diagram of MTCNN is shown in Fig. 2.

The network structure of MTCNN is relatively simple with low computational complexity, enabling real-time detection while ensuring accuracy, which meets the real-time requirements of driver fatigue detection tasks. Experimental results demonstrate that MTCNN provides accurate face landmarks for our fatigue detection algorithm. In future work, we will explore the positive impacts of other face detection algorithms on driver fatigue detection tasks.

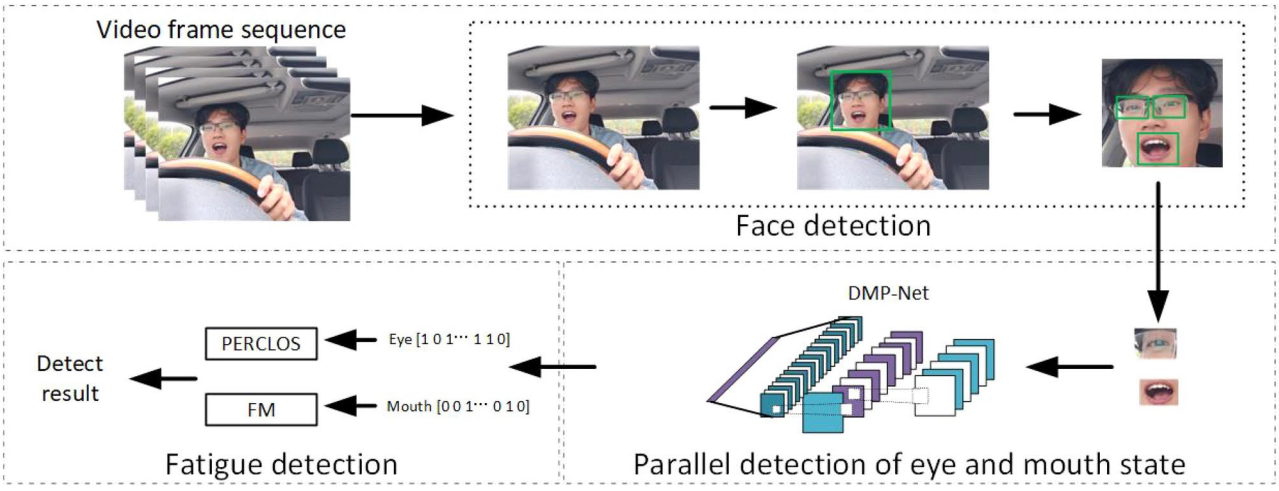

Fig. 1. Overview of the proposed method for driving fatigue detection.

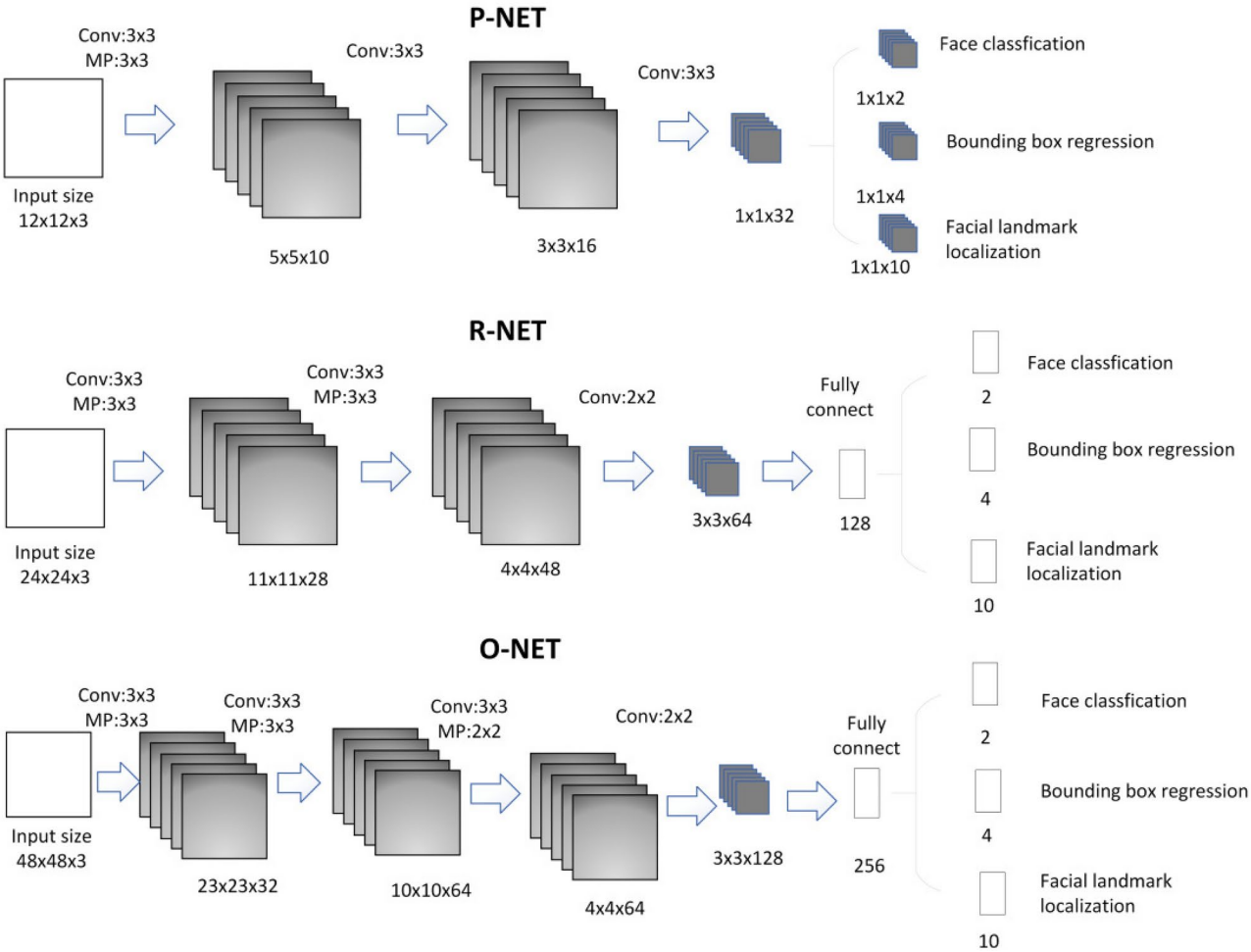

Fig. 2. The structure of the MTCNN, 'Conv' means convolution, 'MP' means max pooling.

The facial expressions of different drivers in a state of fatigue may vary, and features based on a single modality are difficult to adapt to the needs of practical and complex application scenarios, which may lead to the failure of driving fatigue detection models based on a certain modality. Moreover, when drivers lack sufficient rest and feel mentally fatigued, they usually exhibit characteristics such as frequent blinking, prolonged eye closure, yawning. Therefore, we propose to determine driver fatigue based on the state of the driver's eyes and mouth. We design a method to effectively extract the eye and mouth regions from five facial landmarks, as shown in formula (1), where W_e denotes the width of the eye extraction box, H_e denotes the height of the eye extraction box, W_m denotes the width of the mouth extraction box, H_m denotes the height of the mouth extraction box. ER_x and EL_x are the abscissas of the right and left eyes, respectively. MR_x and ML_x are the abscissas of the right and left mouth angles, respectively. Figure 3 shows the extraction result of eyes and mouth area.

$$\begin{cases} W_e = 0.5 * ER_x - EL_x \\ H_e = 0.5 * W_e \\ W_m = MR_x - ML_x \\ H_m = W_m \end{cases} \tag{1}$$

In order to capture the driver's frontal facial image, we install the facial image acquisition device on the center console directly opposite the driver's seat, ensuring that the installation does not interfere with the driver's normal driving. In this scenario, when there are passengers in the back row, the device will capture the facial areas of the rear passengers, thereby interfering with the detection results. To address this, we propose a face detection bounding box filtering method to prevent the detection system from becoming unreliable due to the facial areas of rear passengers.

As shown in Fig. 4, due to the presence of rear passengers, MTCNN detected a series of face detection boxes, where *N* denotes the number of face detection boxes, generally less than or equal to five. The coordinates of the upper left and lower right corner of each face detection box are *x* and *y* respectively. According to formula (2), we get the area of all pixels in each face detection box. Compare the pixel area and threshold of each face detection

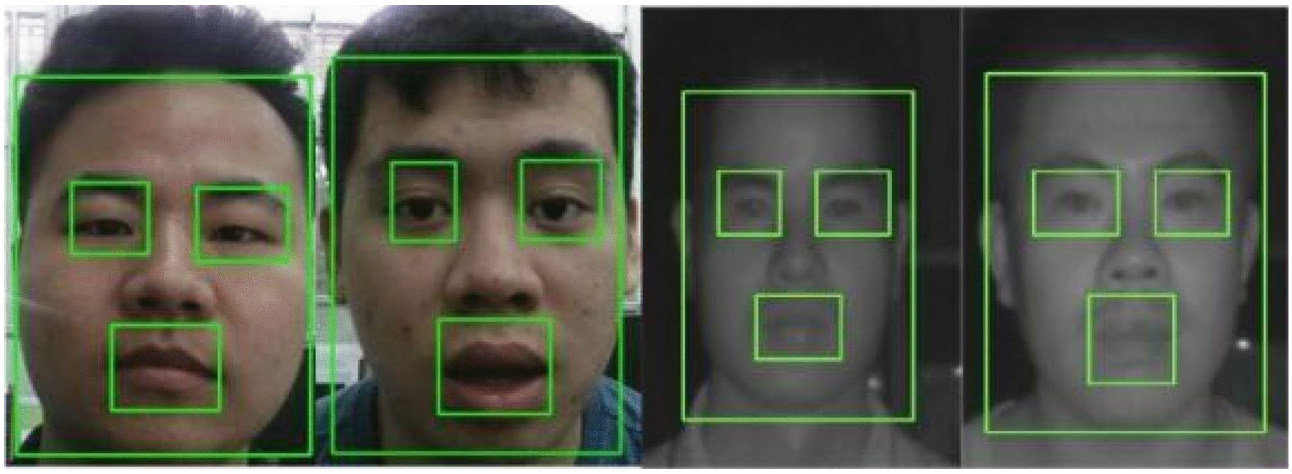

Fig. 3. Infrared camera detects and extracts eye and mouth areas from the left and normal camera from the right.

Fig. 4. Driver face screening based on pixel area size.

box to obtain the threshold for the area of a pixel by formula (3). When the $Np_i < f_{np}$, b_i is thought to be a face detection box for rear passengers.

$$N_p = \sum_{i=0}^{m} \sum_{j=0}^{n} S_{x_i y_j} \quad i = 0, 1, 2 \cdots m, j = 0, 1, 2 \cdots n \tag{2}$$

$$f_{np} = \sum_{i=1}^{n} \frac{N_{p_i}}{n} \quad i = 1, 2 \cdots n \tag{3}$$

Dense multi-pooling convolutional network

Many methods[5–7] have proven that the states of a driver's eyes and mouth are crucial features influencing driving fatigue detection models. Therefore, the DMP-Net is proposed to detect the eye and mouth state in parallel. DMP-Net is implemented based on the dense connection structure of DenseNet[32].

DenseNet has broken the conventional pattern of relying on deepening network layers (such as ResNet) and widening network structures to enhance network performance. Compared to ResNet, DenseNet introduces a more radical dense connectivity mechanism, which interconnects all layers. Specifically, each layer receives all preceding layers as its additional inputs. In ResNet, each layer is shortcut-connected to some preceding layer(s) (typically 2 to 3 layers) through element-wise addition. In DenseNet, however, each layer concatenates with all preceding layers in the channel dimension to serve as the input for the subsequent layer. Moreover, DenseNet

directly concatenates feature maps from different layers, enabling feature reuse and enhancing efficiency. This is particularly important for the real-time requirements of driving fatigue detection tasks. Before inputting the feature maps into the dense structure, a convolution operation is first carried out. Subsequently, two different sizes of average pooling are applied, and the results of these two partial pooling operations are concatenated. The concatenated results are then sent to the subsequent network layers. The concept of this multi-pooling originates from spatial pyramid pooling (SPP)[33], which overcomes the limitation of traditional neural networks on the input image size, effectively extracts spatial features from the image, and can enhance the performance of subsequent models.

The steps of multi-pooling are as follows: first, the original feature maps are obtained from the Conv1 layer. Then proceed with the following operations: (a) the original feature maps are fed into average pooled twice (the pooling size is 2×2, and stride is 1); (b) the original feature maps are fed into another average pooling layer (the pooling size is 3×3, and stride is 1) at the same time; (c) we cascade the two sets of feature maps and input them to latter layers. After the operation of multi-pooling, the output channel is increased.

The structure of our DMP-Net is shown in Fig. 5. It can be seen from the structure diagram that, after the Conv1 layer and multipooling, all convolutional filters are of size 3×3 with a stride of 1. All pooling layers are average pooling. DMP-Net contains a structure similar to a dense block, which has four convolution blocks, blocks and blocks are connected by dense connectivity. The output channels of each convolution block is 12. After convolution, the feature maps are sent to spatial pyramid pooling layer which can convert vectors of different dimensions to output of the same dimension. This allows the network to accept input data of different sizes. After spatial pyramid pooling, the vectors of same dimension are fed into the full connection layer. Finally, a softmax layer divides the outputs into eight categories: the HD color open eye (coe), the HD color close eye (cce), the infrared open eye (ioe), the infrared close eye (ice), the HD color open mouth (com), the HD color close mouth (ccm), the infrared open mouth (iom), the infrared close mouth (icm). An 8-way softmax classification is used as the output layer of DMP-Net. The softmax function is defined as formula (4).

$$p_j = \frac{exp(y'_j)}{\sum_{k=1}^{4} exp(y'_k)}, \quad j = 0, 1, 2, 3, 4, 5, 6, 7 \tag{4}$$

where p_j is the probability of the *jth* class. $y'_j = \sum h_i w_{i,j} + b_j$ represents the outputs of the entire network, while h_i is the outputs of the previous layer, $w_{i,j}$ and b_j are the weight and bias of the layer, respectively.

Loss function: In formula (5), cross-entropy is defined as the loss function.

$$L_n = -\sum_{j=0}^{7} 1\{y = j\} log_{p_j} \tag{5}$$

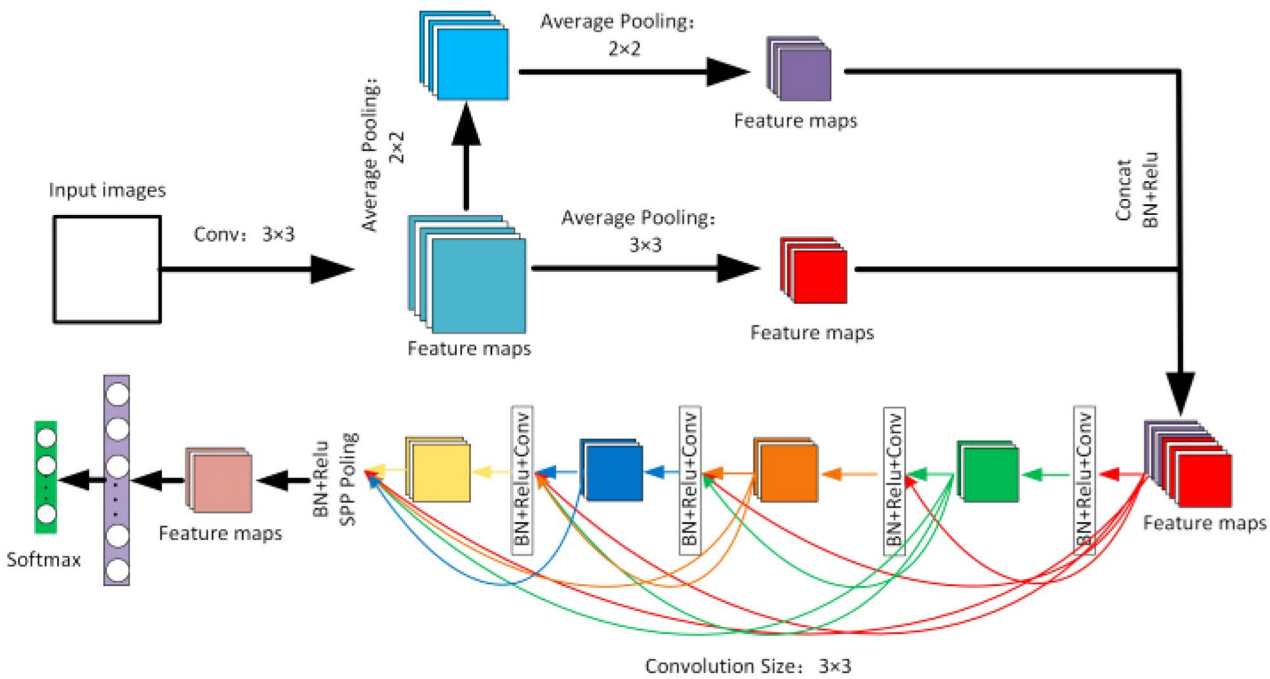

Fig. 5. Architectures of DMP-Net, Conv means convolution.

where L_n is the cross-entropy of the *nth* sample. $1\{y = j\}$ is an characteristic function, which means that if y $\neq$ j, the value is 0, if y = j, the value is 1. Formula (5) is the cross-entropy of a single sample. If there are N training samples, there needs to be an average of the loss function as formula (6).

$$L = \frac{1}{N}\sum_{n=1}^{n} L_n \tag{6}$$

Driving fatigue detection

When the driver feels fatigued, many physiological reactions such as yawning, and prolonged eyes closed. Driver fatigue can be detected by these physiological reactions. In this paper, we adopt the Frequency of Mouth open (FM) parameter and the PERCLOS parameter as indicators to detect driver fatigue. We use the combined states of the eyes and mouth to determine whether a driver is fatigued, which reduces the impact on the detection algorithm's performance when either the eyes or the mouth are obscured or misjudged. In future work, we will explore incorporating features from more modalities to jointly determine driver fatigue, such as changes in the driver's gaze direction and facial expressions, which reflect whether the driver's attention is focused on driving.

Frequency of yawning

The frequency of mouth opening (FM) is the rate at which mouth opening frames occur compared with total frames in a given period. As shown in formula (7).

$$f_{fm} = \frac{n_{mo}}{N_{total}} \times 100\% \tag{7}$$

where n_{mo} represents mouth open frames, while N_{total} represents total frames. Fatigue is more prevalent in drivers with a high value of f_{fm} .

PERCLOS

PERCLOS (percentage of eyelid closure over the pupil over time) is a widely used parameter to detect fatigue. PERCLOS is comparable to FM. It refers to the proportion of frames with closed eyes to all frames in a unit of time. As shown in formula (8).

$$f_{perclos} = \frac{n_{close}}{N_{total}} \times 100\% \tag{8}$$

where N_{total} is the total number of frames, while n_{close} is the number of frames with the eyes closed. When $f_{perclos}$ more than a threshold value (the author in[29] uses 0.15 as the threshold), the driver can be judged as fatigue driving. Fatigue is more prevalent in drivers with a high value of $f_{perclos}$.

In this paper, we use PERCLOS and FM two parameters to judge driver fatigue. The system we designed can be used for real-time detection. The workflow of our system can be divided into four steps:

(1) The video frames are captured by a high-definition camera mounted on the center console in front of the driver.
(2) The pre-trained MTCNN model captures the driver's facial region and segments the eye and mouth areas using five facial landmarks. Our DMP-Net detects the mouth and eye states of each frame. Specifically, Assuming the video segment contains *m* frames. Our method sequentially detects the states of the eyes and mouth in each frame along the temporal dimension and stores the detection results in two separate sequences. As shown in Fig. 6, in these two sequences, we record the detection result of the closed eye and mouth open frames as "1", and the detection results of the open-eye and mouth close frames as "0".
(3) After all video frames have been detected, we determine the values of n_{mo} and n_{close} in Eq. (7) and Eq. (8). The number of "1" in the mouth state result sequence is n_{mo} , and the number of "1" in the eye state sequence is n_{close}. $N_{total} = m$.
(4) Compare f_{fm} and $f_{perclos}$ with their respective predefined thresholds; if they exceed the thresholds, the driver is considered to be fatigued, and a warning is issued.

Experiments

The proposed method is implemented using Pytorch and TensorFlow. All the experiments are conducted on a workstation with a 3.3GHz Intel i7 292 CPU, and 8GB RAM. The first part of our experiment evaluates the speed and accuracy of the DMP network, and the second part is to driver's fatigue detection.

In the face detection section, we use the trained MTCNN model to detect the face and extract eyes and mouth regions. In literature[29], authors use the WIDER FACE dataset[34] to train MTCNN. The WIDER FACE dataset contains 393703 labeled face bounding boxes in 32203 images.

In the first part, we demonstrated the effectiveness of the parallel detection of eye and mouth state based on the ZJU dataset[35], the CEW dataset[36] and our self-collected dataset. In the second part, we have conducted some comparative experiments in the Driving Fatigue Detection module to compare our method and other state-of-the-art methods.

The parameters for training the DMP-Net network model are as follows: the maximum number of iterations is 180,000, the initial learning rate is 0.01, and the learning rate is reduced to 10% of its previous value every 18,000 iterations.

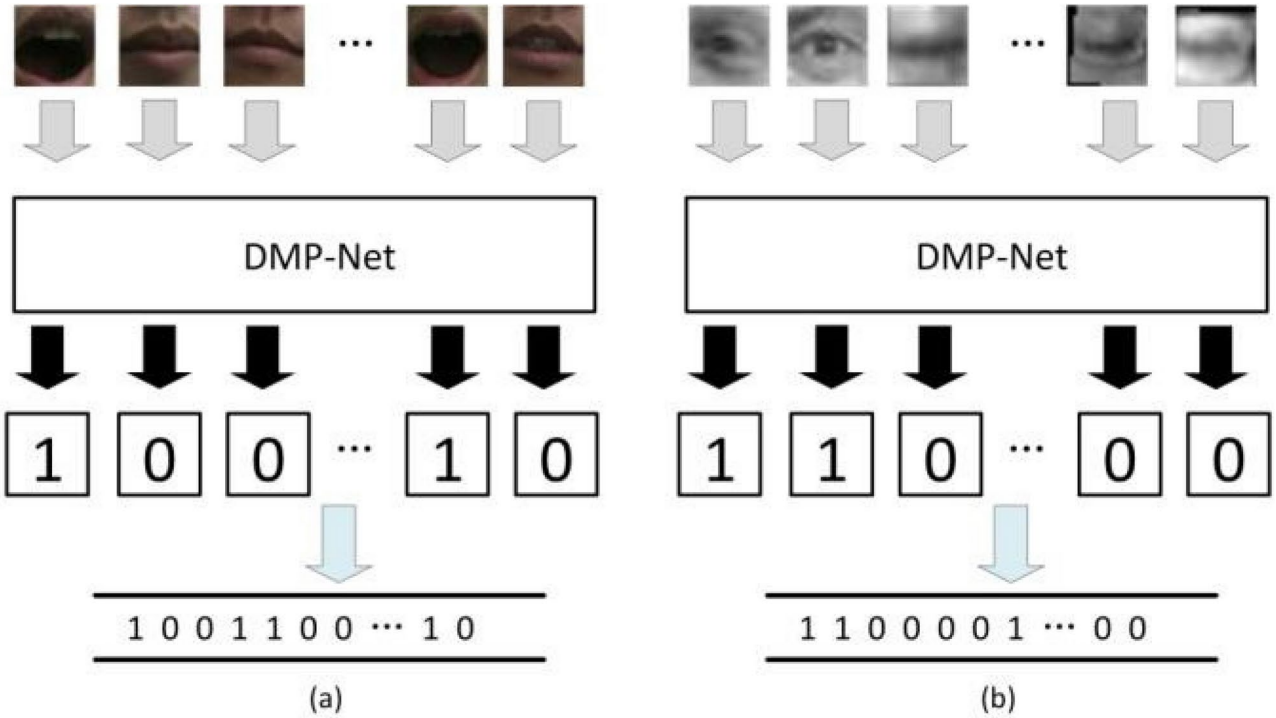

Fig. 6. fatigue detection model. The left side detects the state of the mouth and store in the sequence, and the right side detects the eye state.

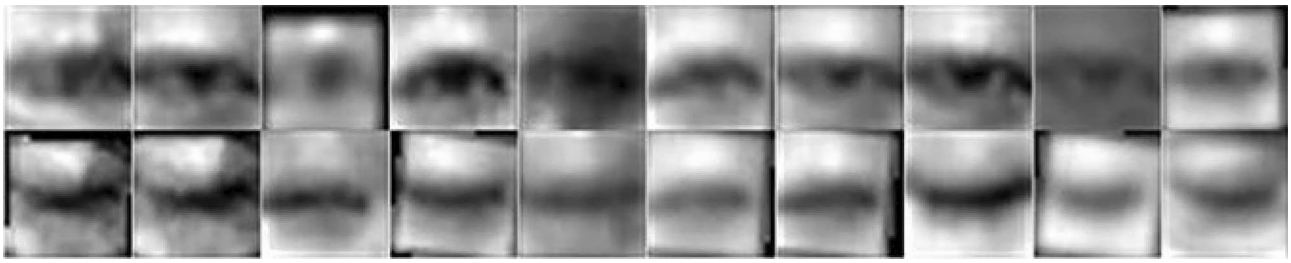

Fig. 7. Sample images from the ZJU Eyeblink dataset.

Dataset

We use the processed ZJU dataset, processed CEW dataset, and our self-collected dataset to train DMP-Net.

ZJU Eyeblink dataset

The ZJU Eyeblink dataset is collected by researchers at Zhejiang University. The dataset consists of 80 video clips from 20 volunteers, 4 clips for each volunteer: one clip for frontal view with glasses, one clip for frontal view without glasses, one clip for frontal view and black frame glasses, and the last clip with an upward view without glasses.

Following[37], there are 6800 open eyes images, of which 1100 are test samples and 5700 are training examples. In total, there are 5820 images with closed eyes, of which 700 are for testing and 5120 are for training (as shown in Fig. 7). The size of images in the ZJU Eyeblink dataset is 24×24.

CEW dataset

The CEW dataset is collected from the Internet and the labeled face in the wild. The CEW dataset consists of 2462 open-eye images and 2384 closed-eye images. (As shown in Fig. 8). Each image size is 24×24. Setting the training set and test set following[37].

self-collected dataset

Our self-collected dataset is collected by the authors. Many public eye datasets are used to recognize eye states. However, no one public mouth state dataset. The purpose of gathering this dataset is to evaluate DMP-Net's effectiveness. Similar to most datasets collected from simulated environments, our dataset also simulates the

Fig. 8. Sample images from the CEW datasets.

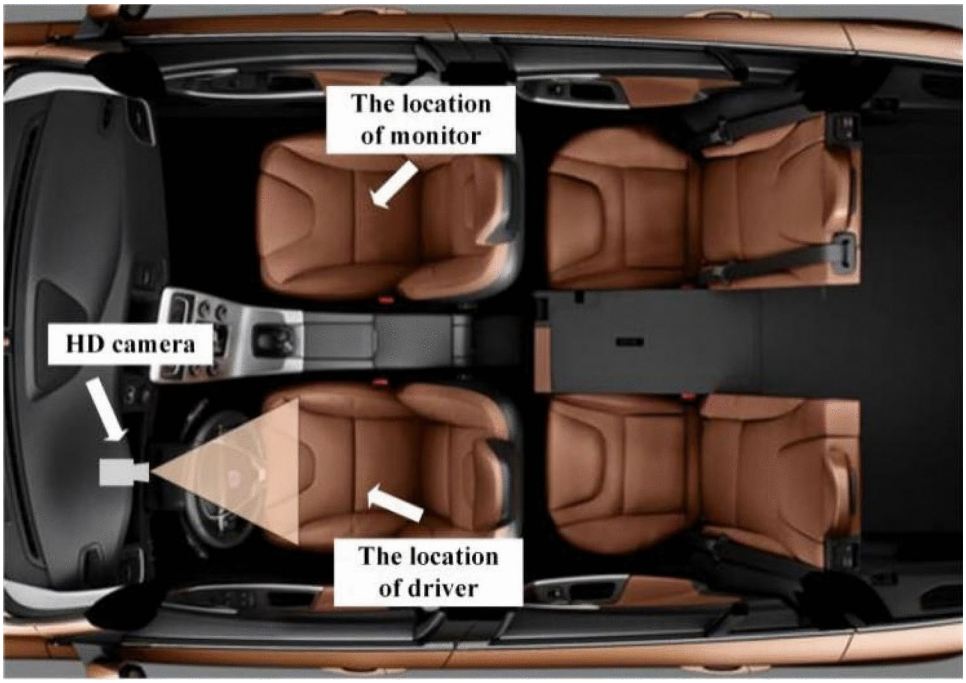

Fig. 9. Diagram of HD camera and driver position.

state of yawning during driving. In future work, we will explore the collection of datasets representing real states. During the data collection process, an HD camera and an infrared camera are used in a driving situation. Twenty volunteers provided 34,325 samples of their eyes and 28,453 samples of their mouths for the dataset. Our collection location was chosen on a wide road with few people because of the variability of the complicated external environment and the risk of fatigued driving. The location of the HD camera and driver is shown in Fig. 9. Images from the frontal view and lateral perspective, both with and without glasses, are among the selections. Finally, the samples will be classified into eight categories (as shown in Fig. 10). Table 2 shows the category and quantity information of the training samples in the self-collected dataset. To balance the samples of fatigue and non-fatigue in the dataset, we collected samples with eyes and mouths open and closed in a nearly 1:1 ratio. In future work, we will further explore balancing samples of different fatigue levels within the dataset.

Parallel detection of eye and mouth state on self-collected datasets

MTCNN has a good performance in face detection. MTCNN is very robust to face detection under different angles and lighting (about 95% validation accuracy in FDDB dataset[38]. MTCNN achieves 99fps in terms of detection speed, with strong real-time performance.

The car will vibrate and bump during driving. The training effect will be poor if the samples that the camera collects are used for direct training. Pre-processing of the original samples is necessary. In order to enhance the details of the picture and to improve the recognition rate, in this paper we first use the image enhancement algorithm based on the histogram equalization[39] to enhance the details of the original frame sequences.

To validate the effectiveness of our method on our self-collected dataset, we contrast DMP-Net with AlexNet[40], ResNet[41], and MSP-Net[37]. Although these methods are relatively traditional, they have been widely accepted and used as benchmarks in the study of Detection of Eye and Mouth State to validate the efficiency of new models. With the continuous development in the field of driving fatigue detection, new methods are constantly emerging. Therefore, in future research, we will compare these newer and more precise methods

Fig. 10. Sample images from the Self-collected dataset. The first two lines' images are captured with a regular camera, while the remaining lines are captured using an infrared camera.

Training object	Category and the number of training samples in self-collected dataset			
Eye	coe	cce	ioe	ice
	11231	11894	6004	5196
Mouth	com	ccm	iom	icm
	9256	6542	5423	7532

Table 2. Category and the quantity of training samples in self-collected dataset. Four categories of eye images: the HD color open eye (coe), the HD color close eye (cce), the infrared open eye (ioe), the infrared close eye (ice). Four categories of mouth images: the HD color open mouth (com), the HD color close mouth (ccm), the infrared open mouth (iom), the infrared close mouth (icm).

Training object	Number of samples			Networks	Accuracy (%)
Eye	Total	training	Testing	AlexNet[40]	95.19
				ResNet[41]	95.79
	34325	32609	1716	MSP-Net[37]	96.06
				Our method	**98.45**
Mouth	Total	training	Testing	AlexNet[40]	96.60
				ResNet[41]	96.48
	31453	29880	1573	MSP-Net[37]	97.86
				Our method	**99.12**

Table 3. Comparison of eye and yawning detection on self-collected datasets. The best performance values are in bold.

to further enhance the quality and accuracy of our work. There are 34,325 eye's samples and 31,453 mouth's samples in our self-collected dataset. About 5% of them are for testing and 95% of them are for training. The experimental results are presented in Table 3, where we can observe that DMP achieves the best performance in the classification of eye and mouth status. In addition, due to the mouth samples are simpler than eye samples (with and without glasses), the lassification accuracy of mouth is slightly higher than eyes. Throughout the experiments, we simulated general driver conditions (both with and without glasses) to obtain results that closely resemble real-world scenarios. Therefore, no independent experiments were conducted solely for drivers with glasses or without glasses.

To ensure that the algorithm we propose can be effectively applied to vehicle-mounted devices with limited resources, we conducted a detailed analysis of the detection speed of DMP-Net, which accounts for the majority of the computational cost of the model. The results are shown in Table 4. From the experimental results, it can be

Networks	Parametes (M)	Time-consuming (ms)
AlexNet[40]	60	32.8
Resnet[41]	1.7	5.6
MSP-Net[37]	-	4.3
Our method	**1.02**	**3.5**

Table 4. Parameter and computational speed comparison of some art-of-the-state models on self-collected datasets. The best results are in bold.

Classifiers	CEW dataset			ZJU dataset		
	ACC (%)	AUC (%)	Time (ms)	ACC (%)	AUC (%)	Time (ms)
LBP+SVM[42]	-	–	-	90.37	-	–
HOG+random ferns[43]	94.15	98.78	4.000	94.76	98.89	9.000
MultiHPOG+SVM[44]	94.45	96.94	38.470	96.83	**99.27**	37.570
MSP-Net[37]	95.36	98.85	1.702	96.89	99.05	1.823
WBCNNTL[45]	–	–	–	97.2	–	–
Burcu et al.[46]	–	–	–	97.3	–	–
DCNNE[47]	**97.59**	–	–	97.99	–	–
Our method	97.23	**99.25**	**1.358**	**98.8**	99.24	**1.156**

Table 5. Comparison of six classifiers on CEW dataset and ZJU dataset. The best results are in bold.

seen that DMP-Net outperforms other models in both accuracy and real-time performance, demonstrating that DMP-Net can meet the real-time requirements of fatigue detection tasks. Table 4 also reports the comparison results of the total number of parameters between our model and other models, demonstrating that our model is more lightweight and can meet the real-time requirements for fatigue detection.

In this paper, we also train and test DMP-Net both on ZJU Eyeblink dataset and CEW dataset. Table 5 shows the experiment results between DMP-Net and several classifiers on these two datasets. We observed that the 'LBP + SVM' classifier has the lowest accuracy rate. The 'MultiHPOG + SVM' classifier outperforms our method on the ZJU dataset in terms of the AUC metric. However, the processing time (37.570ms vs. 1.156ms) is excessive. MSP-Net performs worse than DMP-Net in terms of both accuracy rate and time consumption. The models proposed by Burcu et al.[46], WBCNNTL[45], and DCNNE[47] all perform worse than our method on the ZJU dataset. However, DCNNE achieves slightly higher accuracy than our method on the CEW dataset, which may be attributed to the fact that DCNNE employs an ensemble of two convolutional neural networks (CNNs), each trained to extract eye-related information. This undoubtedly reduces the model's efficiency. By comparison, DMP-Net not only demonstrates superior performance but also boasts impressive detection speed, making it more suitable for integration into driving fatigue detection systems to prevent severe traffic accidents caused by delayed detection.

We utilized three different datasets to validate the performance of our model, and our model exhibited superior performance across all three types of datasets, which also proved that our model does not suffer from overfitting.

Driving fatigue detection

In this paper, the Frequency of Mouth Open (FM) and the PERCLOS are used to measure fatigue. In order to use these two parameters effectively for fatigue detection, reasonable thresholds is required. Geng et al.[48] set PERCLOS threshold to 0.25, while Wan et al[49] use 0.4 as PERCLOS threshold. These instances indicate that different models adopt different thresholds. The PERCLOS and FM thresholds should be adaptively determined through experiments. To ensure that the thresholds we set can be applied to different types of drivers and complex driving environments, we collected videos containing various actions, including blinking, singing, and talking, for experimentation. The experimental results are shown in Fig. 11, the number of false detection (no fatigue but judged as fatigue) and missed detection (fatigue but not detected) at different threshold is obtained.

We choose the threshold corresponding to the point where both the false detection and the missed detection are adequately considered. When the false detection and the missed detection cannot reach the minimum at the same time, we tend to choose the threshold corresponding to the lower point, because missed detection is substantially more expensive than false detection. Therefore, based on the results presented in Fig. 11, we select 0.3 as the threshold for PERCLOS and 0.5 as the threshold for FM.

The pseudocode for the entire architecture of driver fatigue detection is shown in Fig. 12. Classify fatigue levels as follows: (1) When PERCLOS is less than 0.3 and FM is equal or greater than 0.5, it can be regarded as mild fatigue. (2) When PERCLOS is equal or greater than 0.3 and FM is less than 0.5, it can be regarded as moderate fatigue. (3) Severe fatigue may be present when PERCLOS is equal to or higher than 0.3 and FM is equal to or higher than 0.5. These three situations are all fatigue driving.

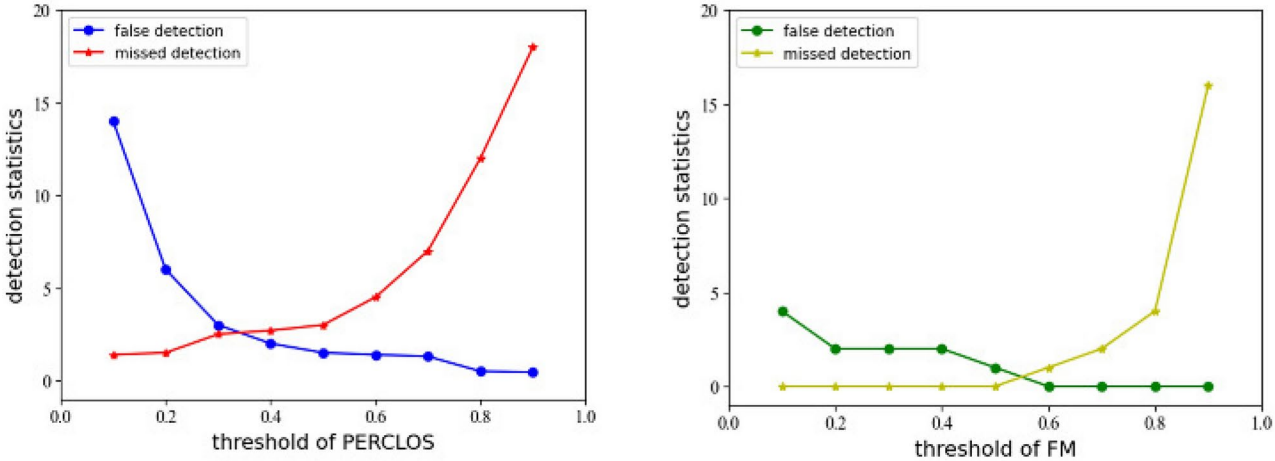

Fig. 11. False and missed detection at various thresholds. The right is for FM, and the left is for PERCLOS. The number of missed detection is shown by the stars, while the number of false detection is represented by the dots.

To verify our method is accurate, we perform an experiment. In this experiment, we recorded the fatigue and non-fatigue status of 20 testers, 10 of whom wore glasses and 10 did not. Each tester was asked to simulate 5 fatigue states and 5 non-fatigue states such as singing, speaking, smiling. The testers' conditions are recorded using both the HD camera and the infrared camera. We use detection precision rate and recall rate to measure the performance of the model. The precision rate is defined as formula (9).

$$P = \frac{n_f - n_m}{n_f - n_m + n_e} \tag{9}$$

where n_f is the overall count of instances of driving fatigue, n_e is the number of false detection and n_m is the number of missed detection. The higher precision is, the higher the classification accuracy of the model.

$$R = \frac{n_f - n_m}{n_f} \tag{10}$$

Lower missed detection probabilities are associated with higher recall levels.

The experimental results presented in Table 6 demonstrate the effectiveness of our method. Specifically, the precision of our method reaches 99.16% and the recall of our method reaches 99.31%.

Conclusion

We propose a new approach to detect driver fatigue based on DSD, DMP-Net and DFD modules. In DSD module, the MTCNN is employed to detect the face landmarks and locate driver's face efficiently. A driver face detection algorithm is also be added to MTCNN, effectively reducing the impact of the back passenger's face on the detection. Then, we propose a novel network, named DMP-Net, which only contains a few parameters and can detect the states of driver's facial action quickly and accurately. Fatigue is judged in DFD modules. The frequency of mouth opening and the percentage of eyelid closure over pupil over time parameters are used in the DFD module to assess driver fatigue using an appropriate threshold.

Our results of the experiments based on benchmarks and our self-collected datasets show that the proposed fatigue driving detection method has a high accuracy. Therefore, our efficient driver fatigue driving detection method has important research significance and is important to improve road safety. Our method is mainly to conduct theoretical research through our self-collected datasets, and only uses the state of eyes and mouth to judge fatigue. In future work, we will consider collecting real driving states of drivers as datasets for training and testing driving fatigue detection models. Additionally, we will take into account the impact of complex factors such as the driver's head position and facial occlusion on model performance, and integrate information from multiple modalities (eyes, mouth, facial expressions, etc.) to enhance the performance of driving fatigue detection.

```
Begin
    Input fixed camera video
    Frames_count :=0
    For frames_count==N_total Do
    Driver's State Detection
    Check states of eyes and mouths
    Load DMP-Net
    Input eye and mouth feature maps F1, F2.
    Parallel Detection of Eye and Mouth State.
    P1 = Pooling(Average, 2x2, Fi)
    P2 = Pooling(Average, 2x2, P1)
    P3 = Pooling(Average, 3x3, P2)
    F' = Concat(P2, P3)
    For i = 1 to 4 Do
        Ci = Conv_block(F', channels=12)
        If i > 1 then
            F' = Concat(F', Ci)
        end if
    end for i
    SPP = SpatialPyramidPool(F', levels=[1,2,4])
    FullyConnected(SPP)
    Load Driving Fatigue Detection
    If PERCLOS < 0.3 and FM >= 0.5 then
    Mild fatigue
    If PERCLOS >= 0.3 and FM < 0.5 then
    Moderate fatigue
    If PERCLOS >= 0.3 and FM >= 0.5 then
    Severe fatigue
End
```

Fig. 12. Pseudocode for the proposed method.

Image type	Total	False detection	Missed detection	Precision (%)	Recall (%)
Normal	200	4	2	99.16	99.31
Infrared	200	12	5	96.23	97.15

Table 6. Results of fatigue detection.

Data availibility

The public datasets CEW and ZJU Eyeblink used in this research can be found at the following links: https://parnec.nuaa.edu.cn/_upload/tpl/02/db/731/template731/pages/xtan/ClosedEyeDatabases.html. and https://blog.csdn.net/weixin_42298876/article/details/112041136. Our self-collected dataset is not publicly available due to the requirement for confidentiality, but is available from the corresponding author on reasonable request.

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References

1. Lin, Z. et al. Fatigue driving recognition based on deep learning and graph neural network. *Biomed. Signal Process. Control* **68**, 102598 (2021).
2. Rather, A. A., Sofi, T. A., Mukhtar, N. A survey on fatigue and drowsiness detection techniques in driving. In *ICCCIS 2021*, 239–244 (IEEE, 2021).
3. Zeller, R., Williamson, A. & Friswell, R. The effect of sleep-need and time-on-task on driver fatigue. *Transp. Res. Part F Traffic Psychol. Behav.* **74**, 15–29 (2020).
4. Sun, W. et al. A real-time fatigue driving recognition method incorporating contextual features and two fusion levels. *IEEE Trans. Intell. Transp. Syst.* **18**, 3408–3420 (2017).
5. Mandal, B. et al. Towards detection of bus driver fatigue based on robust visual analysis of eye state. *IEEE Trans. Intell. Transp. Syst.* **18**, 545–557 (2016).
6. Gu, W. H., Zhu, Y., Chen, X. D., He, L. F. & Zheng, B. B. Hierarchical cnn-based real-time fatigue detection system by visual-based technologies using msp model. *IET Image Process.* **12**, 2319–2329 (2018).
7. Dong, B., Lin, H. & Chang, C. Driver fatigue and distracted driving detection using random forest and convolutional neural network. *Appl. Sci.* **12**, 8674 (2022).
8. Parekh, D. et al. A review on autonomous vehicles: Progress, methods and challenges. *Electronics* **11**, 2162 (2022).
9. State of the art and challenges. Koesdwiady, A., Soua, R. & amp et al. Recent trends in driver safety monitoring systems. *IEEE Trans. Veh. Technol.* **66**, 4550–4563 (2016).
10. Jap, B. T., Lal, S., Fischer, P. & Bekiaris, E. Using eeg spectral components to assess algorithms for detecting fatigue. *Expert Syst. Appl.* **36**, 2352–2359 (2009).
11. Rohit, F. et al. Real-time drowsiness detection using wearable, lightweight brain sensing headbands. *IET Intel. Transport Syst.* **11**, 255–263 (2017).
12. Takei, Y., Furukawa, Y. Estimate of driver's fatigue through steering motion. In *SMC 2005*, vol. 2, 1765–1770 (IEEE, 2005).
13. Sayed, R. & Eskandarian, A. Unobtrusive drowsiness detection by neural network learning of driver steering. *Proc. Inst. Mech. Eng. D J. Automob. Eng.* **215**, 969–975 (2001).
14. Zhong, Y., Du, L., Zhang, K., Sun, X. Localized energy study for analyzing driver fatigue state based on wavelet analysis. In *ICWAPR 2007*, vol. 4, 1843–1846 (IEEE, 2007).
15. Wang, Z., Hong, Q. & Wang, X. A memristive circuit implementation of eyes state detection in fatigue driving based on biological long short-term memory rule. *IEEE/ACM Trans. Comput. Biol. Bioinform.* **18**, 2218–2229 (2020).
16. Fu, R., Wang, H. & Zhao, W. Dynamic driver fatigue detection using hidden markov model in real driving condition. *Expert Syst. Appl.* **63**, 397–411 (2016).
17. Zhang, C., Wang, H. & Fu, R. Automated detection of driver fatigue based on entropy and complexity measures. *IEEE Trans. Intell. Transp. Syst.* **15**, 168–177 (2013).
18. Pavel, M. I., Tan, S. Y. & Abdullah, A. Vision-based autonomous vehicle systems based on deep learning: A systematic literature review. *Appl. Sci.* **12**, 6831 (2022).
19. Min, W. et al. Traffic sign recognition based on semantic scene understanding and structural traffic sign location. *IEEE Trans. Intell. Transp. Syst.* **23**, 15794–15807 (2022).
20. Yang, H. et al. Driver yawning detection based on subtle facial action recognition. *IEEE Trans. Multimedia* **23**, 572–583 (2020).
21. Wang, Q. et al. Inter-domain adaptation label for data augmentation in vehicle re-identification. *IEEE Trans. Multimedia* **24**, 1031–1041 (2021).
22. Chang, T., Chen, Y. Driver fatigue surveillance via eye detection. In *ITSC 2014*, 366–371 (IEEE, 2014).
23. Dinges, D. F., Grace, R. Perclos: A valid psychophysiological measure of alertness as assessed by psychomotor vigilance. *US Dep. Transp. Fed. Highw. Adm. Publ. FHWA-MCRT-98-006* (1998).
24. Li, L., Xie, M., Dong, H. A method of driving fatigue detection based on eye location. In *ICCSN 2011*, 480–484 (IEEE, 2011).
25. Ibrahim, M. M. et al. Yawn analysis with mouth occlusion detection. *Biomed. Signal Process. Control* **18**, 360–369 (2015).
26. Zhang, Y. & Hua, C. Driver fatigue recognition based on facial expression analysis using local binary patterns. *Optik* **126**, 4501–4505 (2015).
27. Park, S., Pan, F., Kang, S., Yoo, C. D. Driver drowsiness detection system based on feature representation learning using various deep networks. In *ACCV 2016 Workshops*, 154–164 (Springer, 2017).
28. Zhao, L., Wang, Z., Wang, X. & Liu, Q. Driver drowsiness detection using facial dynamic fusion information and a dbn. *IET Intell. Transp. Syst.* **12**, 127–133 (2018).
29. Zhang, K. et al. Joint face detection and alignment using multitask cascaded convolutional networks. *IEEE Signal Process. Lett.* **23**, 1499–1503 (2016).
30. Min, W. et al. Real-time face recognition based on pre-identification and multi-scale classification. *IET Comput. Vis.* **13**, 165–171 (2019).
31. Neubeck, A., Van Gool, L. Efficient non-maximum suppression. In *ICPR 2006*, vol. 3, 850–855 (IEEE, 2006).
32. Huang, G., Liu, Z., et al. Densely connected convolutional networks. In *CVPR 2017*, 4700–4708 (2017).
33. He, K. et al. Spatial pyramid pooling in deep convolutional networks for visual recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* **37**, 1904–1916 (2015).
34. Song, F. et al. Eyes closeness detection from still images with multi-scale histograms of principal oriented gradients. *Pattern Recognit.* **47**, 2825–2838 (2014).
35. Pan, G., Sun, L., Wu, Z., Lao, S. Eyeblink-based anti-spoofing in face recognition from a generic webcamera. In *ICCV 2007*, 1–8 (IEEE, 2007).
36. Song, F., Tan, X., Liu, X. & Chen, S. Eyes closeness detection from still images with multi-scale histograms of principal oriented gradients. *Pattern Recognit.* **47**, 2825–2838 (2014).
37. Gu, W. H. et al. Hierarchical cnn-based real-time fatigue detection system by visual-based technologies using msp model. *IET Image Process.* **12**, 2319–2329 (2018).
38. Jain, V., Learned-Miller, E. Fddb: A benchmark for face detection in unconstrained settings. Tech. Rep., UMass Amherst Tech. Rep. (2010).

39. Ibrahim, H. & Kong, N. S. P. Brightness preserving dynamic histogram equalization for image contrast enhancement. *IEEE Trans. Consum. Electron.* **53**, 1752–1758 (2007).
40. Krizhevsky, A., Sutskever, I. & Hinton, G. E. Imagenet classification with deep convolutional neural networks. *Commun. ACM* **60**, 84–90 (2017).
41. He, K., Zhang, X., et al. Deep residual learning for image recognition. In *CVPR 2016*, 770–778 (2016).
42. Wu, Y., Lee, T., et al. An eye state recognition method for drowsiness detection. In *VTC 2010*, 1–5 (IEEE, 2010).
43. Dong, Y. et al. Comparison of random forest, random ferns and support vector machine for eye state classification. *Multimed. Tools Appl.* **75**, 11763–11783 (2016).
44. Song, F. et al. Eyes closeness detection from still images with multi-scale histograms of principal oriented gradients. *Pattern Recognit.* **47**, 2825–2838 (2014).
45. Liu, Z.-T. et al. Eye state detection based on weight binarization convolution neural network and transfer learning. *Appl. Soft Comput.* **109**, 107565 (2021).
46. Kır Savaş, B., Becerikli, Y. Driver fatigue detection via eye state analyses based on deep learning approach. In *Proc. of the International Conference on Smart City Applications*, 452–462 (Springer, 2022).
47. Saurav, S., Gidde, P., Saini, R. & Singh, S. Real-time eye state recognition using dual convolutional neural network ensemble. *J. Real-Time Image Proc.* **19**, 607–622 (2022).
48. Geng, L. et al. Driver fatigue detection method based on facial behavior analysis. *Comput. Eng.* **44**, 274–279 (2018).
49. Wan, Y. & Xie, J. One algorithm of driver fatigue detecting based on perclos. *Agric. Equip. Technol.* **35**, 25–28 (2009).

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Declarations

Competing interests

The authors declare no competing interest.

Additional information

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